

Lab Experiments 13-15

Data Handling & Visualization: Python + R, Code & Output

Each experiment: Python code + output, followed by R code + output

Date: 22/07/2026

Experiment 13

Geographic Data

PYTHON

Code

```
import matplotlib.pyplot as plt
import pandas as pd

df = pd.DataFrame({
    'City': ['City A', 'City B', 'City C'],
    'Population': [500000, 700000, 600000],
    'AvgTemperature': [75, 68, 80],
    'Elevation': [1000, 800, 1200]
})

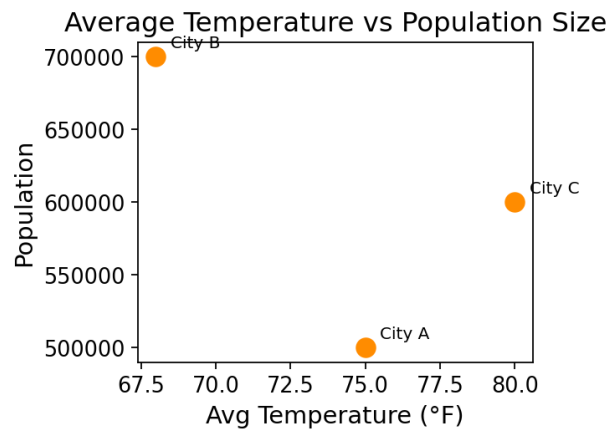
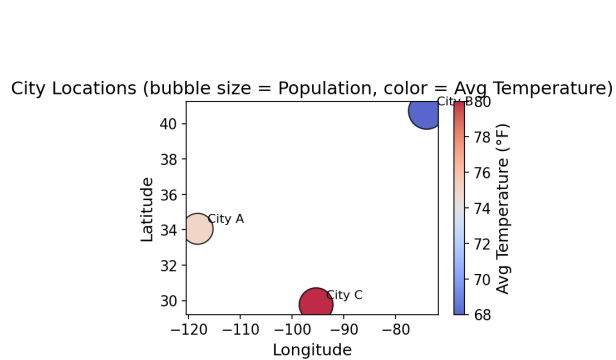
# No latitude/longitude is given for these cities, so plausible coordinates
# are assumed here to render a map-style bubble chart.
df['Latitude'] = [34.05, 40.71, 29.76]
df['Longitude'] = [-118.24, -74.01, -95.37]

# 1. Map chart - cities plotted by location, bubble size = population
plt.figure(figsize=(7, 5))
sc = plt.scatter(df['Longitude'], df['Latitude'], s=df['Population'] / 800,
                 c=df['AvgTemperature'], cmap='coolwarm', edgecolor='black', alpha=0.85)
for _, row in df.iterrows():
    plt.annotate(row['City'], (row['Longitude'], row['Latitude']),
                 textcoords="offset points", xytext=(8, 5), fontsize=10)
plt.colorbar(sc, label="Avg Temperature (°F)")
plt.title("City Locations (bubble size = Population, color = Avg Temperature)")
plt.xlabel("Longitude"); plt.ylabel("Latitude")
plt.tight_layout()
plt.show()

# 2. Scatter plot - average temperature vs population size
plt.figure(figsize=(7, 5))
plt.scatter(df['AvgTemperature'], df['Population'], s=100, color='darkorange')
for _, row in df.iterrows():
    plt.annotate(row['City'], (row['AvgTemperature'], row['Population']),
                 textcoords="offset points", xytext=(8, 5), fontsize=9)
plt.title("Average Temperature vs Population Size")
plt.xlabel("Avg Temperature (°F)"); plt.ylabel("Population")
plt.tight_layout()
plt.show()
# Insight: in this small sample, the warmest city (City C) also has a mid-range
# population - no strong linear trend is visible with only 3 points.

# 3. Table - geographic data per city
fig, ax = plt.subplots(figsize=(7, 2))
ax.axis('off')
display_df = df[['City', 'Population', 'AvgTemperature', 'Elevation']]
tbl = ax.table(cellText=display_df.values, colLabels=display_df.columns, loc='center', cellLoc='center')
tbl.auto_set_font_size(False)
tbl.set_fontsize(11)
tbl.scale(1, 1.8)
plt.title("City Geographic Data", pad=20)
plt.tight_layout()
plt.show()
```

Output



City Geographic Data

City	Population	AvgTemperature	Elevation
City A	500000	75	1000
City B	700000	68	800
City C	600000	80	1200

R

Code

```
library(ggplot2)
library(maps)
library(gridExtra)
library(grid)

df <- data.frame(
  City = c('City A', 'City B', 'City C'),
  Population = c(500000, 700000, 600000),
  AvgTemperature = c(75, 68, 80),
  Elevation = c(1000, 800, 1200)
)

# No latitude/longitude is given for these cities, so plausible coordinates
# are assumed here to render a map-style bubble chart.
df$Latitude <- c(34.05, 40.71, 29.76)
df$Longitude <- c(-118.24, -74.01, -95.37)

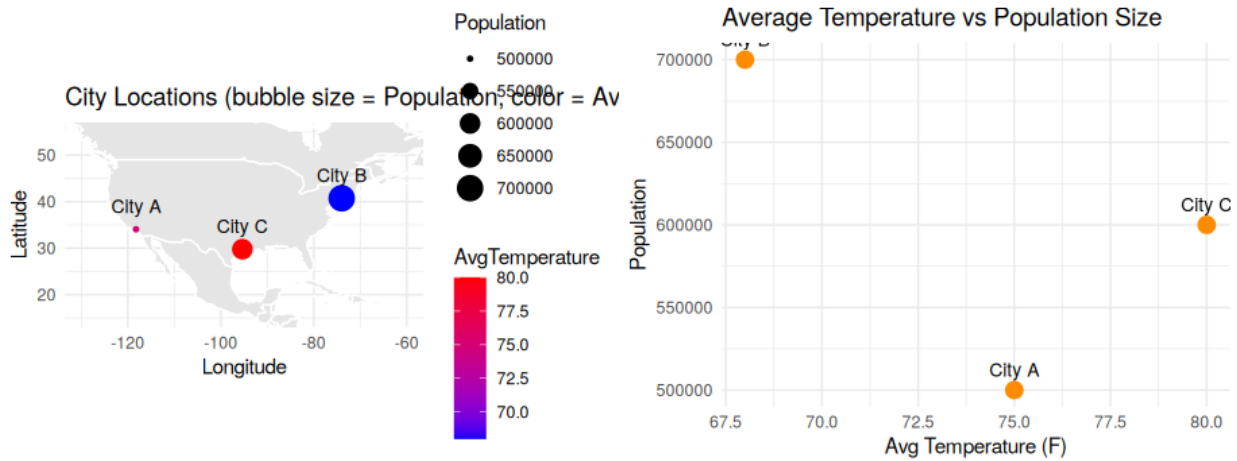
# 1. Map chart - cities plotted on a world map, bubble size = population
world <- map_data("world")
p1 <- ggplot() +
  geom_polygon(data=world, aes(x=long, y=lat, group=group), fill="gray90", color="white") +
  geom_point(data=df, aes(x=Longitude, y=Latitude, size=Population, color=AvgTemperature)) +
  geom_text(data=df, aes(x=Longitude, y=Latitude, label=City), vjust=-1.2, size=3.5) +
  scale_color_gradient(low="blue", high="red") +
  coord_fixed(xlim=c(-130, -60), ylim=c(15, 55)) +
  labs(title="City Locations (bubble size = Population, color = Avg Temperature)",
        x="Longitude", y="Latitude") +
  theme_minimal()
print(p1)

# 2. Scatter plot - average temperature vs population size
p2 <- ggplot(df, aes(x=AvgTemperature, y=Population)) +
  geom_point(size=4, color="darkorange") +
  geom_text(aes(label=City), vjust=-1, size=3.5) +
  labs(title="Average Temperature vs Population Size", x="Avg Temperature (F)", y="Population") +
  theme_minimal()
```

```
print(p2)
# Insight: in this small sample, the warmest city (City C) also has a mid-range
# population - no strong linear trend is visible with only 3 points.

# 3. Table - geographic data per city
grid.newpage()
grid.table(df[, c('City', 'Population', 'AvgTemperature', 'Elevation')], rows=NULL)
```

Output



City	Population	AvgTemperature	Elevation
City A	5e+05	75	1000
City B	7e+05	68	800
City C	6e+05	80	1200

Experiment 14

Survey Results

PYTHON

Code

```
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np

df = pd.DataFrame({
    'Respondent': [1, 2, 3],
    'Question1': ['A', 'B', 'C'],
    'Question2': ['B', 'A', 'A'],
    'Question3': ['C', 'D', 'B']
})

# 1. Stacked bar chart - distribution of answers for Question1
q1_counts = df['Question1'].value_counts().sort_index()
plt.figure(figsize=(6, 5))
bottom = 0
for answer, count in q1_counts.items():
    plt.bar('Question1', count, bottom=bottom, label=answer)
    bottom += count
plt.title("Distribution of Answers for Question 1 (Stacked Bar)")
plt.ylabel("Number of Responses")
plt.legend(title="Answer")
plt.tight_layout()
plt.show()

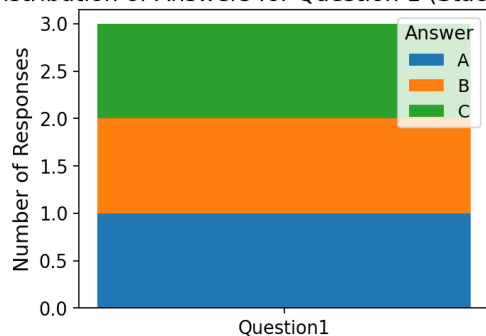
# 2. Radar chart - overall response pattern across all three questions
# Radar axes = questions; each respondent's answer is mapped to a numeric code
# (A=1, B=2, C=3, D=4) purely to plot it spatially - the letters themselves
# have no inherent order, so this is a positional/pattern view, not a ranking.
answer_map = {'A': 1, 'B': 2, 'C': 3, 'D': 4}
questions = ['Question1', 'Question2', 'Question3']
angles = np.linspace(0, 2 * np.pi, len(questions), endpoint=False).tolist()
angles += angles[:1]

fig, ax = plt.subplots(figsize=(6, 6), subplot_kw=dict(polar=True))
for _, row in df.iterrows():
    values = [answer_map[row[q]] for q in questions]
    values += values[:1]
    ax.plot(angles, values, marker='o', label=f"Respondent {row['Respondent']}")
    ax.fill(angles, values, alpha=0.1)
ax.set_xticks(angles[:-1])
ax.set_xticklabels(questions)
ax.set_yticks([1, 2, 3, 4])
ax.set_yticklabels(['A', 'B', 'C', 'D'])
ax.set_title("Response Pattern Across Questions (Radar Chart)")
ax.legend(loc='upper right', bbox_to_anchor=(1.3, 1.1))
plt.tight_layout()
plt.show()

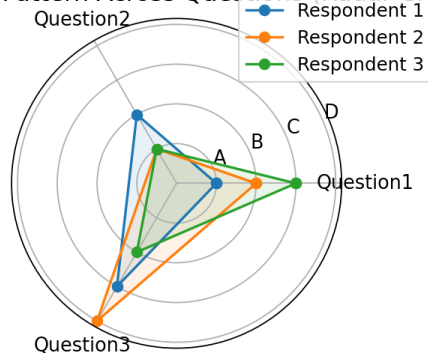
# 3. Table - survey response data per respondent
fig, ax = plt.subplots(figsize=(6, 2))
ax.axis('off')
tbl = ax.table(cellText=df.values, colLabels=df.columns, loc='center', cellLoc='center')
tbl.auto_set_font_size(False)
tbl.set_fontsize(11)
tbl.scale(1, 1.8)
plt.title("Survey Response Data by Respondent", pad=20)
plt.tight_layout()
plt.show()
```

Output

Distribution of Answers for Question 1 (Stacked Bar)



Response Pattern Across Questions (Radar Chart)



Survey Response Data by Respondent

Respondent	Question1	Question2	Question3
1	A	B	C
2	B	A	D
3	C	A	B

R

Code

```
library(ggplot2)
library(dplyr)
library(gridExtra)
library(grid)

df <- data.frame(
  Respondent = 1:3,
  Question1 = c('A', 'B', 'C'),
  Question2 = c('B', 'A', 'A'),
  Question3 = c('C', 'D', 'B')
)

# 1. Stacked bar chart - distribution of answers for Question1
q1_counts <- as.data.frame(table(df$Question1))
colnames(q1_counts) <- c('Answer', 'Count')
q1_counts$Question1 <- "Question1"
p1 <- ggplot(q1_counts, aes(x=Question, y=Count, fill=Answer)) +
  geom_col() +
  labs(title="Distribution of Answers for Question 1 (Stacked Bar)", y="Number of Responses") +
  theme_minimal()
print(p1)

# 2. Radar chart - overall response pattern across all three questions
# Radar axes = questions; each respondent's answer is mapped to a numeric code
# (A=1, B=2, C=3, D=4) purely to plot it spatially - the letters themselves
# have no inherent order, so this is a positional/pattern view, not a ranking.
# Built manually with coord_polar since the fmsb radar-chart package is not
# installable in this environment (no CRAN access).
answer_map <- c(A=1, B=2, C=3, D=4)
long_df <- df %>%
  tidyr::pivot_longer(cols=c(Question1, Question2, Question3),
    names_to="Question", values_to="Answer") %>%
  mutate(Value = answer_map[Answer])

p2 <- ggplot(long_df, aes(x=Question, y=Value, group=factor(Respondent), color=factor(Respondent))) +
  geom_polygon(fill=NA, linewidth=1) +
  geom_point(size=2) +
  coord_polar() +
```

```
scale_y_continuous(breaks=1:4, labels=c('A', 'B', 'C', 'D'), limits=c(0,4)) +
labs(title="Response Pattern Across Questions (Radar Chart)", color="Respondent", y="", x="") +
theme_minimal()
print(p2)
```

```
# 3. Table - survey response data per respondent
grid.newpage()
grid.table(df, rows=NULL)
```

Output



Respondent	Question1	Question2	Question3
1	A	B	C
2	B	A	D
3	C	A	B

Experiment 15

Product Sales Analysis (Monthly)

PYTHON

Code

```
import matplotlib.pyplot as plt
import pandas as pd

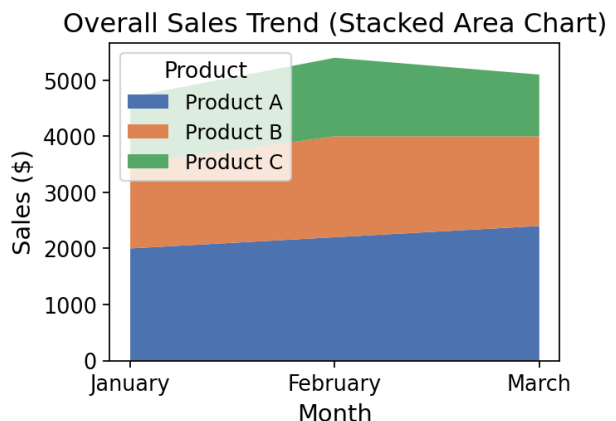
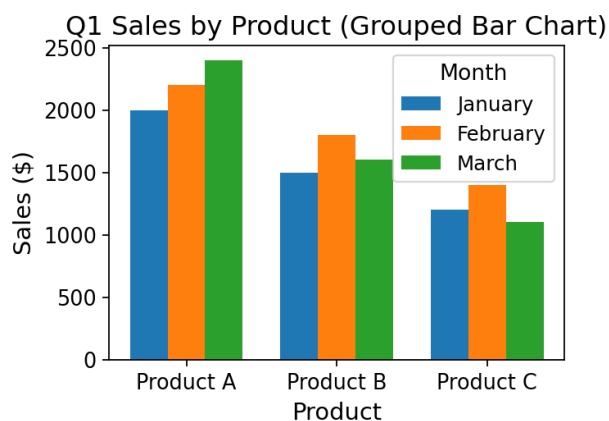
# Note: this dataset and task set is identical to Experiment 6 (Monthly Product
# Sales) in the given LOE - reproduced here as its own experiment per the sheet.
df = pd.DataFrame({
    'ProductID': [1, 2, 3],
    'ProductName': ['Product A', 'Product B', 'Product C'],
    'January': [2000, 1500, 1200],
    'February': [2200, 1800, 1400],
    'March': [2400, 1600, 1100]
})

# 1. Grouped bar chart - Q1 sales per product
months = ['January', 'February', 'March']
x = range(len(df))
width = 0.25
plt.figure(figsize=(7, 5))
for i, m in enumerate(months):
    plt.bar([p + i * width for p in x], df[m], width=width, label=m)
plt.xticks([p + width for p in x], df['ProductName'])
plt.title("Q1 Sales by Product (Grouped Bar Chart)")
plt.xlabel("Product"); plt.ylabel("Sales ($)")
plt.legend(title="Month")
plt.tight_layout()
plt.show()

# 2. Stacked area chart - overall sales trend across products
plt.figure(figsize=(7, 5))
plt.stackplot(months, df.set_index('ProductName')[months].values,
              labels=df['ProductName'], colors=['#4C72B0', '#DD8452', '#55A868'])
plt.title("Overall Sales Trend (Stacked Area Chart)")
plt.xlabel("Month"); plt.ylabel("Sales ($)")
plt.legend(title="Product", loc='upper left')
plt.tight_layout()
plt.show()

# 3. Table - monthly sales data
fig, ax = plt.subplots(figsize=(7, 2.2))
ax.axis('off')
tbl = ax.table(cellText=df.values, colLabels=df.columns, loc='center', cellLoc='center')
tbl.auto_set_font_size(False)
tbl.set_fontsize(11)
tbl.scale(1, 1.8)
plt.title("Monthly Sales Data by Product", pad=20)
plt.tight_layout()
plt.show()
```

Output



Monthly Sales Data by Product

ProductID	ProductName	January	February	March
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

R

Code

```
library(ggplot2)
library(reshape2)
library(gridExtra)
library(grid)

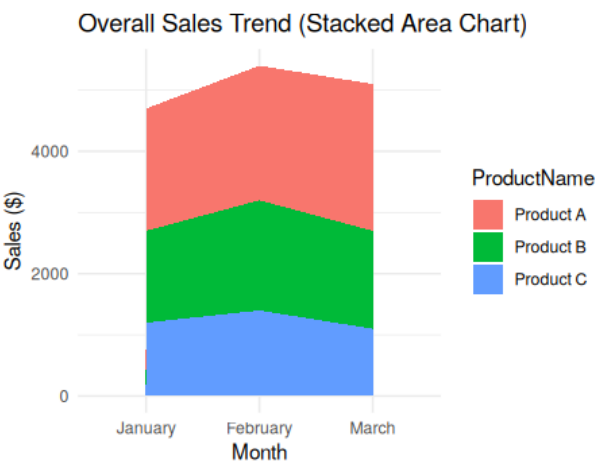
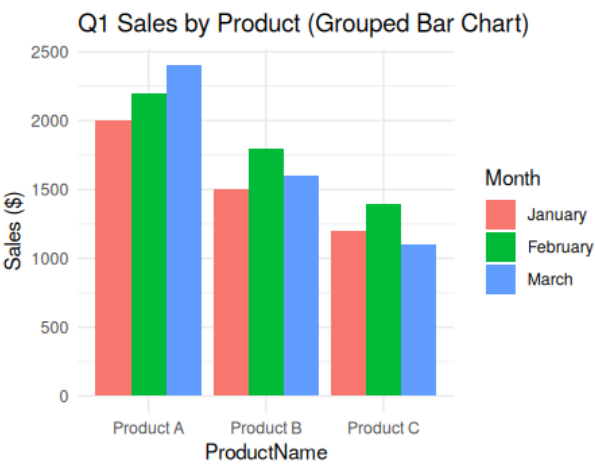
# Note: this dataset and task set is identical to Experiment 6 (Monthly Product
# Sales) in the given LOE - reproduced here as its own experiment per the sheet.
df <- data.frame(
  ProductID = 1:3,
  ProductName = c('Product A', 'Product B', 'Product C'),
  January = c(2000, 1500, 1200),
  February = c(2200, 1800, 1400),
  March = c(2400, 1600, 1100)
)

# 1. Grouped bar chart - Q1 sales per product
long_df <- melt(df, id.vars=c('ProductID', 'ProductName'), variable.name='Month', value.name='Sales')
long_df$Month <- factor(long_df$Month, levels=c('January', 'February', 'March'))
p1 <- ggplot(long_df, aes(x=ProductName, y=Sales, fill=Month)) +
  geom_col(position=position_dodge()) +
  labs(title="Q1 Sales by Product (Grouped Bar Chart)", y="Sales ($)") +
  theme_minimal()
print(p1)

# 2. Stacked area chart - overall sales trend across products
p2 <- ggplot(long_df, aes(x=Month, y=Sales, fill=ProductName, group=ProductName)) +
  geom_area() +
  labs(title="Overall Sales Trend (Stacked Area Chart)", y="Sales ($)") +
  theme_minimal()
print(p2)

# 3. Table - monthly sales data
grid.newpage()
grid.table(df, rows=NULL)
```

Output



ProductID	ProductName	January	February	March
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100